🗺 The Longitude — FULL MIND MAP

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Chapter 02_ii_The_Longitude

Topic Index

Longitude-Time Relationship — Base Formula

- **1° longitude = 4 minutes time** (15° longitude = 1 hour) — yeh saari calculation ka base formula hai. *[Q3,Q26]*
- Cairo Greenwich se 2 hours aage hai → Cairo **~30° East** pe hai (2 hours × 15° = 30°, East kyunki time aage hai). *[Q1]*
- 90° longitude difference = **6 hours** time difference (90×4=360 min = 6 hrs). *[Q3]*
- Greenwich noon (12) pe agar kisi jagah 5pm ho, wo jagah **75° East** hogi (5hrs × 15° = 75°, East kyunki aage hai). *[Q4]*

Longitude Calculation Problems (Numericals)

- 82°30′E pe noon ho, toh 6:30am kahan hoga? Answer: **0° (Prime Meridian)** — 5.5 hrs difference = 82°30′ longitude difference, jo exactly 82°30′E se subtract karke 0° deta hai. *[Q2]*
- Greenwich se telegram 12 noon pe bheja, 12 min transmission, 6pm pahuncha → destination **87° East** (348 min effective time × 15/60 = 87°). *[Q14]*
- IST meridian (82°30′E) pe noon ho, kisi jagah 6am ho (breakfast time) → us jagah ka longitude **7°30′ West** hoga (6 hrs = 90° difference, IST se west mein 90° → 82°30′E - 90° = 7°30′W). *[Q13]*
- GMT 3am ho aur local time 6am ho, toh place **45° East** pe hoga (3 hrs ahead = 45° East). *[Q24]*
- Greenwich noon ho, Thimphu (90°E) ka local time **6pm** hoga (90° = 6 hrs ahead). *[Q25]*

> ⚠ **EXAM TRAP**
> • Longitude-time problems mein hamesha check karo: East = time AAGE, West = time PEECHHE. Direction galat lena sabse common mistake hai. *[Q1,Q4,Q24,Q25]*

Great Circle aur Prime Meridian

- 0° aur **180°** longitude milkar ek Great Circle banate hain (Prime Meridian + its opposite meridian). *[Q5]*

New Millennium First Sunrise (India)

- New Millennium (2000) mein India mein sunrise ki first ray **Katchal Island** (Andaman-Nicobar) ke paas dekhi gayi — closest option **93°30′E**. *[Q6]*

Standard Time — Basis

- Standard time ka basis **Longitude** hai — country ek standard meridian choose karta hai, uska local mean time standard time banta hai. *[Q7]*

Prime Meridian — Countries se Guzarna

- Prime Meridian in countries se guzarti hai: UK, France, Spain, Algeria, Mali, Burkina Faso, Togo, Ghana. *[Q8,Q9]*
- Prime Meridian **Niger, Nigeria, Portugal** se NAHI guzarti. *[Q8,Q9]*

GMT-Same-Time Places

- GMT/UTC use karne wale places (is question context mein): Accra (Ghana), Dublin (Ireland), Lisbon (Portugal) — **Madrid** Central European Time use karta hai, GMT nahi. *[Q10]*
- London, Lisbon, Accra — teeno same GMT time share karte hain; **Addis Ababa (Ethiopia)** alag hai — GMT+3 use karta hai (38°45′E pe located). *[Q11]*

GMT Se Aage/Peeche — Countries Ranking

- Standard time offsets: Japan = +9, Iraq = +3, Greece = +2, Cuba = −5, Costa Rica = −6. Order (most-ahead to most-behind): **Japan** → **Iraq** → **Greece** → **Cuba** → **Costa Rica**. *[Q12]*

International Date Line (IDL)

- **IDL** ~180° meridian pe **Pacific Ocean** se guzarti hai — political/administrative convenience ke liye kahin-kahin mudi hui hai. *[Q15]*
- IDL aur Greenwich ke beech time difference **12 hours** hai (180° × 4 min = 720 min = 12 hrs). *[Q16]*
- IDL ke around minimum fixed distance NAHI hoti (kyunki IDL kai jagah bent/turned hai, e.g. Alaska ke paas) — baaki meridians (Prime, 45°N/S) mein aisi problem nahi. *[Q17]*
- IDL ke sabse close strait: **Bering Strait (~168°23′W)** — Malacca, Florida, Gibraltar straits IDL se bahut door hain. *[Q19]*

IDL Crossing — Date Change Puzzles

- Agar Earth ka rotation direction REVERSE ho jaaye, aur IDL pe 12 noon ho, toh IST kya hoga? Answer: **18:30 hrs** — rotation reverse hone se longitude-time relation bhi reverse ho jaata; jahan IST normally GMT se aage hota, wahan peeche ho jaata (180° meridian aur Greenwich ke beech 12-hour gap ko naye direction mein apply karke 18:30 milta hai). *[Q18]*
- Ship Aleutian region mein (jahan IDL bent hai) 180° meridian cross karta hai 1 Jan 23:30 pe — actual Date Line cross NAHI karta (deviation ki wajah se), toh 1 hour baad diary mein **2 January, 00:30** likhega. *[Q20]*
- Ship 90°W pe Monday 10am se westward 15°/hour speed se chalta hai — IDL cross karne se THEEK PEHLE local solar time **Monday 10:00am hi rahega** (speed effect aur time-shift cancel out ho jaate hain); IDL cross karne ke baad calendar **Tuesday** ho jaayega. *[Q21]*
- Anadyr (Siberia) aur Nome (Alaska) geographically close hain lekin IDL region ke opposite sides pe — different calendar days ho sakte hain (Statement I correct); lekin 'Anadyr Monday = Nome Tuesday' wala specific claim **galat** hai (Statement II incorrect). *[Q22]*

> ⚠ **EXAM TRAP**
>
> • IDL crossing problems mein carefully dekho ki ship **actual bent Date Line** cross kar raha hai ya sirf 180° meridian — dono alag ho sakte hain kuch regions (jaise Aleutian/Alaska) mein. *[Q20]*

Shortest Distance — Great Circle Route

- 2 places ke beech Earth surface pe sabse kam distance wala route **Great Circle Route** hota hai — ships/aircraft isi ka use karte hain jahan practical ho. *[Q23]*

Natural vs Man-made Time Units

- Standard Time NAHI hai natural time unit (yeh human convention hai, ek chosen meridian ke local mean time pe based) — Day, Lunar month, Tropical year sab natural units hain. *[Q26]*

Time Zones — Countries with Dependencies

- Dependencies/overseas territories count karte hue: UK=9, Denmark=5, New Zealand=5, Australia=9, Brazil=4 time zones. Jinke **4 se zyada** time zones hain: UK, Denmark, New Zealand, Australia (4 countries) — Brazil exclude hota hai. *[Q27]*